

Cricbuzz Match Data Analysis using SQL, Python & Streamlit

1. Introduction

This project focuses on analyzing cricket match data collected from Cricbuzz and transforming it into meaningful insights using data analysis techniques. The goal is to convert raw match data into an interactive dashboard that allows users to explore team performance, match frequency, and relationships between teams.

The project integrates **SQL for data storage and querying**, **Python for data processing**, **Streamlit for dashboard creation**, and a **basic Machine Learning model** to enhance analytical depth.

2. Problem Statement

Raw cricket match data is not directly useful for analysis or decision-making. It lacks structure, visualization, and interpretability.

The problem addressed in this project is:

- How to store and manage match data efficiently
 - How to analyze team participation and match distribution
 - How to present insights in an interactive and user-friendly way
 - How to enhance analysis using basic machine learning
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3. Objectives

- Import and store cricket match data in a structured database
 - Perform data analysis using SQL and Python
 - Build an interactive dashboard using Streamlit
 - Generate insights from match data
 - Apply Machine Learning to identify patterns in team activity
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4. Tools & Technologies Used

- **Python** – Data processing and application logic
 - **PostgreSQL** – Database management
 - **Streamlit** – Dashboard development
 - **Pandas** – Data manipulation
 - **Scikit-learn** – Machine Learning (KMeans Clustering)
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5. Data Collection

The dataset was collected in CSV format containing match details such as:

- Match ID
- Match name
- Team 1
- Team 2

The dataset was imported into PostgreSQL for structured storage.

- CSV file preview

	A	B	C
1	Match	Team1	Team2
2	Scotland vs Oman	Scotland	Oman
3	Indonesia vs Sweden	Indonesia	Sweden
4	Indonesia vs Sweden	Indonesia	Sweden
5	France vs Norway	France	Norway
6	Portugal vs Norway	Portugal	Norway
7	Portugal vs France	Portugal	France
8	Portugal vs Norway	Portugal	Norway
9	France vs Norway	France	Norway
10	Rajasthan Royals vs Mumbai Indians	Rajasthan Royals	Mumbai Indians
11	Delhi Capitals vs Gujarat Titans	Delhi Capitals	Gujarat Titans
12	Hyderabad Kingsmen vs Peshawar Zalmi	Hyderabad Kingsmen	Peshawar Zalmi
13	Lahore Qalandars vs Islamabad United	Lahore Qalandars	Islamabad United
14	Sri Lanka A vs New Zealand A	Sri Lanka A	New Zealand A
15	Sri Lanka Women U19 vs Australia Women U19	Sri Lanka Women U19	Australia Women U19
16	Zambia Women vs Lesotho Women	Zambia Women	Lesotho Women
17	Mozambique Women vs Brazil Women	Mozambique Women	Brazil Women
18	Botswana Women vs Malawi Women	Botswana Women	Malawi Women
19	Brazil Women vs Zambia Women	Brazil Women	Zambia Women
20	Lesotho Women vs Mozambique Women	Lesotho Women	Mozambique Women
21	Botswana Women vs Zambia Women	Botswana Women	Zambia Women
22	Mozambique Women vs Malawi Women	Mozambique Women	Malawi Women
23	Brazil Women vs Lesotho Women	Brazil Women	Lesotho Women
24	Malawi Women vs Zambia Women	Malawi Women	Zambia Women

- Table created in pgAdmin

```
21 --QUERY 1 - Portugal matches
22 ▾ SELECT *
23 FROM matches
24 WHERE team1 = 'Portugal' OR team2 = 'Portugal';
25
26 --QUERY 2 - Rajasthan Royals
27 ▾ SELECT *
28 FROM matches
29 WHERE team1 = 'Rajasthan Royals' OR team2 = 'Rajasthan Royals';
30
31 --QUERY 3 - Brazil Women
32 ▾ SELECT *
33 FROM matches
34 WHERE team1 = 'Brazil Women' OR team2 = 'Brazil Women';
35
36 --QUERY 4 - Indonesia
37 ▾ SELECT *
38 FROM matches
39 WHERE team1 = 'Indonesia' OR team2 = 'Indonesia';
40
```

6. Data Storage & SQL Processing

A table named matches was created in PostgreSQL to store the dataset.

Key steps performed:

- Table creation
- Data import using COPY command
- Query execution for validation

SQL was used to:

- Retrieve match data
- Filter matches based on teams
- Perform grouping operations
- SQL table data

- Query execution result

41	--SORTING DATA (ORDER BY)
42	--QUERY 1 -- SORT BY TEAM NAME
43	SELECT *
44	FROM matches
45	ORDER BY team1;
46	
47	--QUERY 2 -- DESCENDING ORDER
48	SELECT *
49	FROM matches
50	ORDER BY team1 DESC;
51	
52	--QUERY 3 -- MOST MATCHES BY TEAM
53	SELECT team1, COUNT(*)

	id [PK] integer	match text	team1 text	team2 text
1	24			
2	20	Botswana Women vs Zambia Women	Botswana Women	Zambia Women
3	17	Botswana Women vs Malawi Women	Botswana Women	Malawi Women
4	22	Brazil Women vs Lesotho Women	Brazil Women	Lesotho Women
5	18	Brazil Women vs Zambia Women	Brazil Women	Zambia Women
6	10	Delhi Capitals vs Gujarat Titans	Delhi Capitals	Gujarat Titans
7	4	France vs Norway	France	Norway
8	8	France vs Norway	France	Norway
9	11	Hyderabad Kingsmen vs Peshawar Zalmi	Hyderabad Kingsmen	Peshawar Zalmi
10	2	Indonesia vs Sweden	Indonesia	Sweden
11	3	Indonesia vs Sweden	Indonesia	Sweden
12	12	Lahore Qalandars vs Islamabad United	Lahore Qalandars	Islamabad United
13	19	Lesotho Women vs Mozambique Women	Lesotho Women	Mozambique Women
14	23	Malawi Women vs Zambia Women	Malawi Women	Zambia Women
15	21	Mozambique Women vs Malawi Women	Mozambique Women	Malawi Women
16	16	Mozambique Women vs Brazil Women	Mozambique Women	Brazil Women

7. Data Preprocessing

Basic preprocessing steps were applied:

- Ensured correct column mapping
- Removed inconsistencies in data
- Structured dataset for analysis

The dataset was kept simple but clean to allow efficient querying and visualization.

8. Streamlit Dashboard

An interactive dashboard was built using Streamlit.

Features:

- Display complete match dataset
- Filter matches by team
- Advanced filtering (Team1 vs Team2)

- Display filtered results dynamically

This allows users to explore data without writing queries.

- Full dashboard view

Cricbuzz Match Analysis

This dashboard shows match data scraped from Cricbuzz website

Match Data

	id	match	team1	team2
11	12	Lanore Qalandars vs Islamabad United	Lanore Qalandars	Islamabad United
12	13	Sri Lanka A vs New Zealand A	Sri Lanka A	New Zealand A
13	14	Sri Lanka Women U19 vs Australia Women U	Sri Lanka Women U19	Australia Women U19
14	15	Zambia Women vs Lesotho Women	Zambia Women	Lesotho Women
15	16	Mozambique Women vs Brazil Women	Mozambique Women	Brazil Women
16	17	Botswana Women vs Malawi Women	Botswana Women	Malawi Women
17	18	Brazil Women vs Zambia Women	Brazil Women	Zambia Women
18	19	Lesotho Women vs Mozambique Women	Lesotho Women	Mozambique Women
19	20	Botswana Women vs Zambia Women	Botswana Women	Zambia Women
20	21	Mozambique Women vs Malawi Women	Mozambique Women	Malawi Women
21	22	Brazil Women vs Lesotho Women	Brazil Women	Lesotho Women

Select Team

Scotland

Filtered Data

	id	match	team1	team2
0	1	Scotland vs Oman	Scotland	Oman

Select Team 1

Scotland

Select Team 2

Oman

Filtered Match Data

	id	match	team1	team2
0	1	Scotland vs Oman	Scotland	Oman

Total Matches

24

9. Data Visualization

Visualizations were added to improve understanding of data patterns.

Charts included:

- Matches per Team (Bar Chart)
- Matches between Teams (Bar Chart)

Insights:

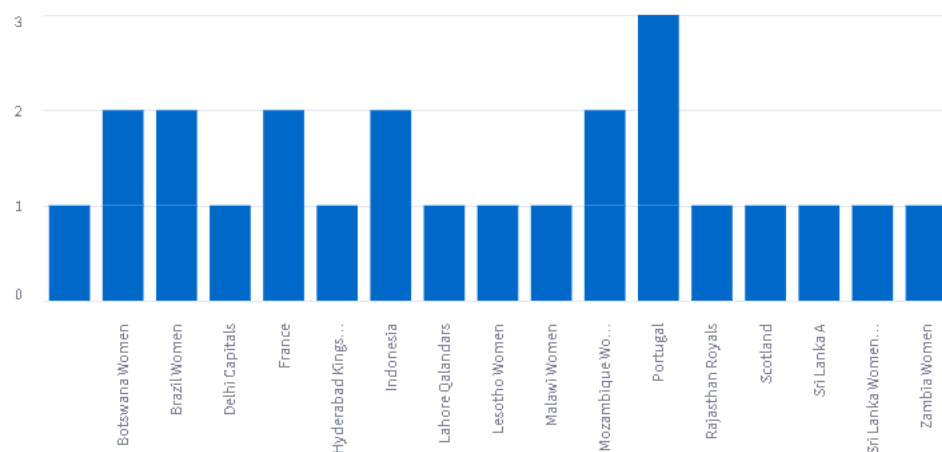
- Some teams appear more frequently than others

- Certain team combinations occur repeatedly
- Charts from dashboard

Matches per Team



Matches between Teams



10. CRUD Operations

The application supports basic database operations:

- **Create** – Add new match records
- **Read** – View match data
- **Update** – Modify existing records
- **Delete** – Remove records

This simulates real-world database interaction.

- CRUD functionality in app

Add New Match

Match (Team1 vs Team2)

India vs Pakistan

Team 1

India

Team 2

Pakistan

Add Match

Match added successfully!

Delete Match

Select Match to Delete

Scotland vs Oman



Delete Match

11. Machine Learning Implementation

To enhance the project and address analytical depth, a **KMeans Clustering model** was implemented.

Objective:

Group teams based on number of matches played.

Process:

- Count matches per team
- Apply KMeans clustering ($k = 3$)
- Categorize teams into:
 - High activity teams
 - Medium activity teams
 - Low activity teams

Insights:

- Some teams are highly active and appear frequently
- Others have lower participation

- Clustering helps identify patterns not visible in raw data
- Cluster output table
- ML visualization

Team Clustering (ML)

	Team	Matches	Cluster
0	Portugal	3	2
1	Indonesia	2	1
2	France	2	1
3	Mozambique Women	2	1
4	Botswana Women	2	1
5	Brazil Women	2	1
6	Scotland	1	0
7	Rajasthan Royals	1	0
8	Delhi Capitals	1	0
9	Hyderabad Kingsmen	1	0

12. Key Insights

- Total matches analyzed: 23
- Certain teams dominate match frequency
- Repeated match combinations exist
- Team activity varies significantly
- Machine learning reveals grouping patterns among teams

13. Limitations

- Dataset is small and limited in scope
 - No historical or performance-based metrics included
 - Analysis is limited to match occurrence only
 - No deep statistical testing performed
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14. Future Improvements

- Use larger and more detailed datasets
 - Add match outcomes and player statistics
 - Implement advanced Machine Learning models
 - Deploy Streamlit app online
 - Improve dashboard storytelling and UI
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15. Conclusion

This project successfully demonstrates an end-to-end data analysis workflow:

- Data collection and storage
- SQL-based querying
- Python-based analysis
- Interactive dashboard creation
- Basic Machine Learning integration

The project converts raw cricket data into actionable insights and provides an interactive platform for exploration.